Real Estate Buyer Segmentation & Investment Profiling

Machine Learning Project Report | Synthetic Demonstration Dataset

Executive summary. This project demonstrates an end-to-end customer segmentation workflow for real estate market intelligence. A reproducible synthetic dataset was generated, cleaned, explored, encoded, scaled and clustered using K-Means, with hierarchical clustering used as a comparison. The dashboard supports filtering and exploratory analysis.

Data integrity notice: All records in this project are synthetic and were created for learning and demonstration. They are not actual customers, transactions, or proprietary Parcl data.

1. Business problem and objectives

Real estate organizations serve buyers with different goals, financing needs, geographies and purchasing capacities. A single generic marketing approach can reduce relevance. The project objective is to identify behaviorally similar buyer groups and translate their characteristics into actionable, testable marketing hypotheses.

Objectives: (1) prepare a reliable analytical dataset; (2) profile demographic, geographic and financing patterns; (3) identify clusters; (4) compare cluster quality; (5) present findings through an interactive dashboard; and (6) document limitations and recommendations.

2. Dataset and feature dictionary

The raw dataset contains 1,512 rows, including 12 intentional duplicate client records. After duplicate removal, 1,500 unique records remain. The original brief's fields are included, with three explicitly synthetic extensions: income_bracket, property_budget_usd and units_interested. Age is derived from date_of_birth.

Field	Meaning	Type
client_id	Unique synthetic buyer identifier	Identifier
client_type	Individual or Corporate	Categorical
gender	Self-reported gender category	Categorical
country	Country of residence	Categorical
region	Broad geographic region	Categorical
date_of_birth	Date of birth used to derive age	Date
acquisition_purpose	Investment, Personal Use, or Business Use	Categorical
loan_applied	Whether buyer applied for financing	Binary
referral_channel	Customer acquisition source	Categorical
satisfaction_score	Satisfaction rating from 1 to 5	Numeric
income_bracket	Synthetic income category; added to support profiling	Ordinal categorical
property_budget_usd	Synthetic indicative property budget in USD	Numeric
units_interested	Number of units buyer is interested in	Numeric
age	Derived age at 2026-11-04, included in cleaned dataset	Numeric
cluster	K-Means cluster identifier	Model output
buyer_segment	Descriptive label derived from cluster profile	Model interpretation

3. Data preparation

Duplicate client_id values were removed. Missing gender and referral_channel values were filled with their respective modes; missing satisfaction_score values were filled with the median. date_of_birth was parsed and converted to age using 4 November 2026 as the reference date. The original date field was removed from the model input after deriving age. A production workflow should also check invalid dates, impossible ages, inconsistent category spelling and outlier sensitivity.

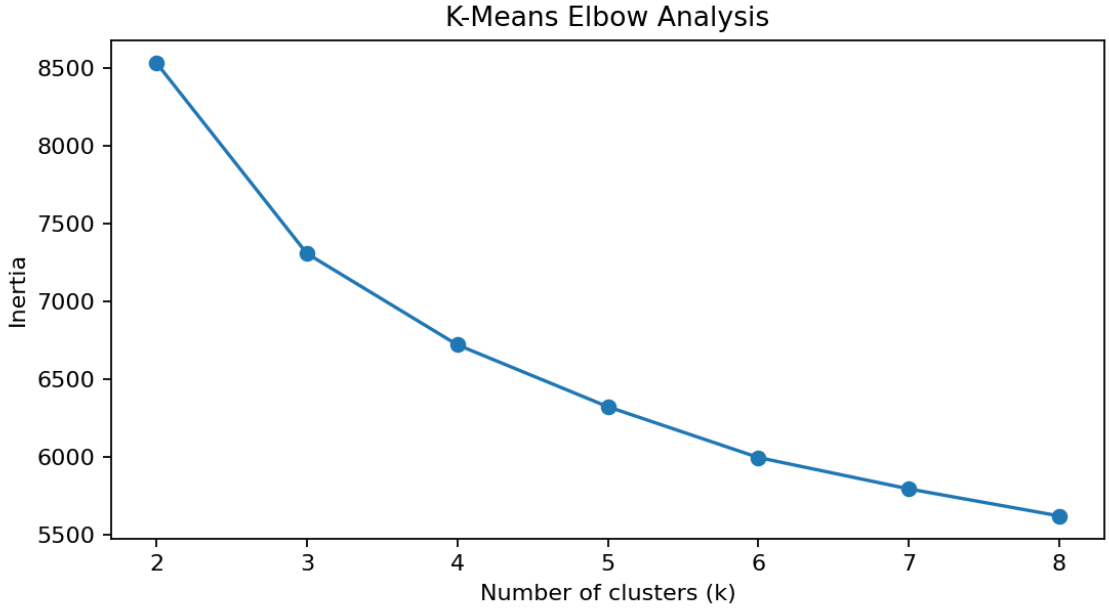
4. Exploratory analysis

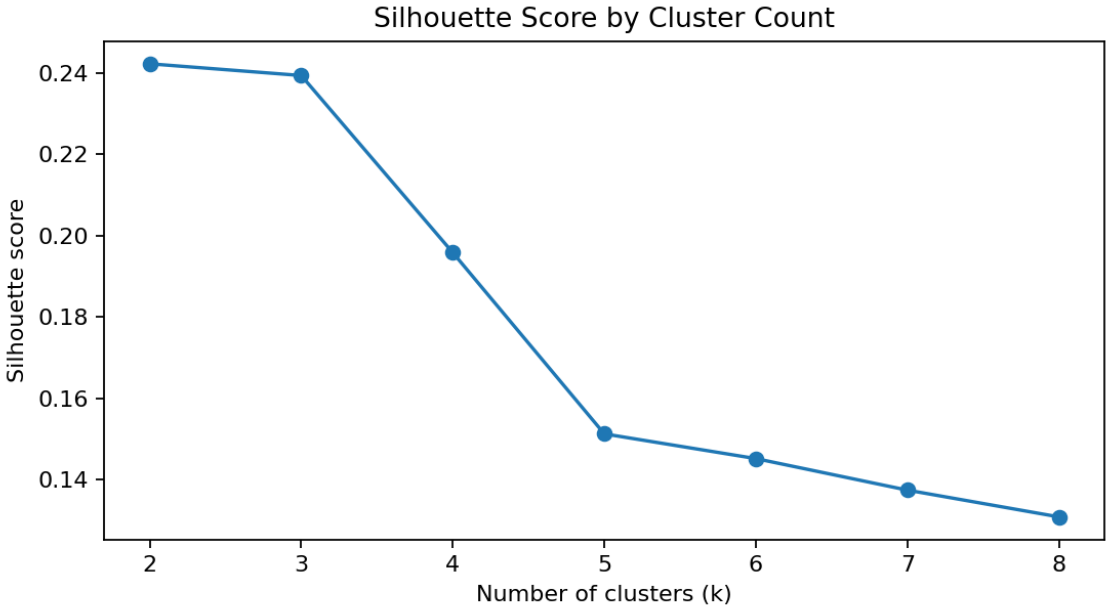
Exploratory analysis examines buyer counts, purpose, client type, region, financing, satisfaction, budget and units of interest. The charts in the visuals folder provide model-selection and segment summaries. The synthetic data generator intentionally creates differentiated patterns, so these findings are demonstrations rather than market estimates.

5. Modeling methodology

Categorical variables (client_type, region, acquisition_purpose, loan_applied, referral_channel and income_bracket) are one-hot encoded. Numeric variables (age, satisfaction_score, property_budget_usd and units_interested) are standardized. K-Means is evaluated for k values 2 through 8 using inertia and silhouette score. The highest silhouette score is used as the selected k. Agglomerative hierarchical clustering with Ward linkage is compared on a reproducible sample of up to 500 records.

Selected K-Means k: **2**. K-Means silhouette: **0.242**. Hierarchical sample silhouette: **0.248**. Silhouette values closer to 1 indicate better separation; values near 0 indicate overlapping clusters. These values suggest modest separation and should not be overstated.



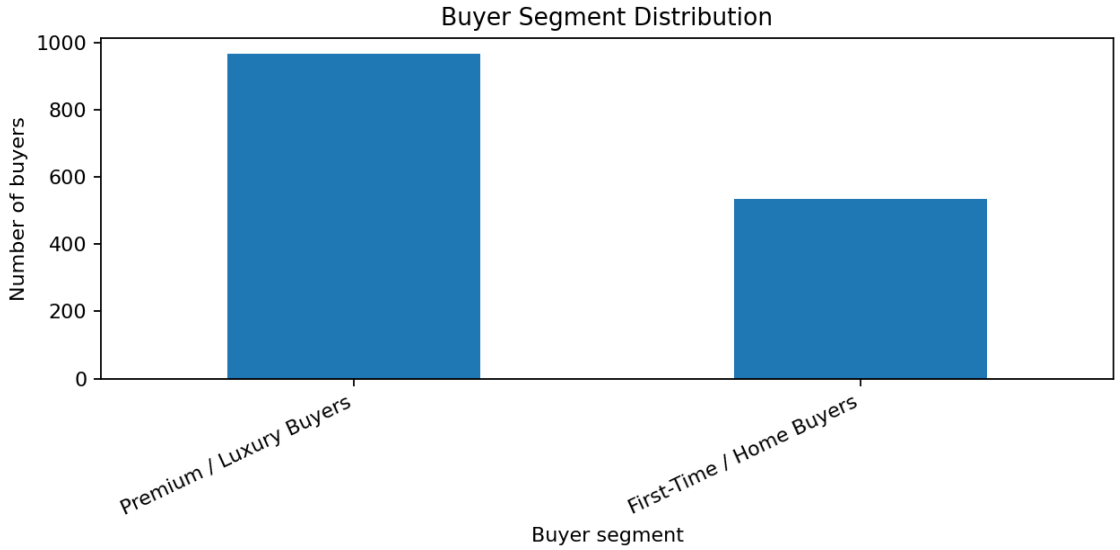


6. Cluster interpretation

The selected model produced two broad groups in this synthetic run. The first is a higher-budget, older, investment-oriented group with lower loan usage and more units of interest. The second is a younger,

lower-budget, personal-use group with substantially higher loan usage. Labels are assigned from observed profiles and are descriptive, not ground-truth identities.

Cluster	Buyers	Avg age	Investment share	Loan share	Avg budget (USD)	Avg units
0	965	47.5	83.0%	22.0%	$488,679	2.47
1	535	30.9	4.0%	78.0%	$71,074	1.01



7. Business recommendations

For the higher-budget investment-oriented group, test premium property content, portfolio-level consultations and cross-border market information where legally appropriate. For the younger personal-use group, test first-home education, transparent affordability tools and financing guidance. Compare campaign conversion, qualified lead rate, cost per qualified lead and customer satisfaction before reallocating budget. Avoid using protected or sensitive attributes for discriminatory targeting.

8. Dashboard and deployment

The Streamlit application provides country, region, acquisition-purpose and client-type filters; KPI summaries; segment distribution; regional purpose comparison; loan mix; budget distribution; segment profile table; and CSV export. Run with: streamlit run app/streamlit_app.py.

9. Limitations, ethics and next steps

The dataset is synthetic and cannot establish real market behavior. Income and property budget are synthetic extensions rather than fields in the original project brief. K-Means results depend on feature design, scaling, encoding and random initialization. Silhouette alone does not prove business usefulness. Before production: obtain consented representative records, assess missingness and drift, test multiple feature sets and cluster stability, conduct fairness/privacy review, document retention controls, and validate recommendations with business stakeholders.

10. Conclusion

This project provides a reproducible demonstration of a machine-learning buyer segmentation workflow and an interactive analytics interface. The current generated sample yields two broad behavioral clusters with modest separation. The work is suitable as a learning prototype; real business decisions require validation on authentic, governed customer data.